Chemical Equilibrium Lectures

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Electrolytes

Conducting an electric current

3 classes of electrolytes acids, bases, and salts.

Strong electrolytes consist of solutes that dissociate completely in solution.

Strong acids, strong bases, and soluble salts

Weak electrolytes consist of solutes that dissociate only a little in solution.

Weak acids, weak bases, and slightly soluble salts

An **acid**:

Mineral acids can be defined as a substance **that produces hydrogen ions, H^+**, in aqueous solutions.

$$HCl + H_2O \rightarrow H_3O^+ (aq) + Cl^- (aq)$$

Organic acids can often be recognized by **the presence of the COOH group** in the formula.

Weak acids ionize only slightly in dilute aqueous solution.

$$CH_3COOH(aq) + H_2O \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$$

A **base** is:

a substance **that produces hydroxide ions**, OH^-, in aqueous solutions.

A **salt** is:

a compound that **contains a cation and an anion**

4-5

Nonelectrolytes are substances that not produce ions in solution, and therefore are considered nonconductors (gases)

In a liquid-vapor equilibrium, for example, the processes are **evaporation and condensation.**

in which **CH_3COOH** and **H_2O** are reacting at the same rate as **H_3O^+** and **$C_2H_3O_2^-$**.

Recall that such **an equilibrium is established when two oposing processes occur at equal rates**.

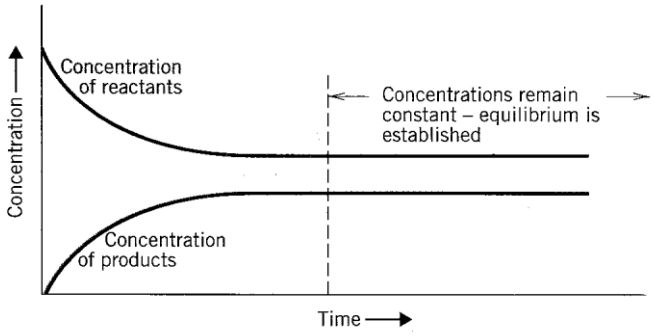
When acetic acid is added to the water, the forward reaction (ionization)occurs rapidly at first, but then slows as the number of $HC_2H_3O_2$ molecules decreases.

As the concentrations of the ions increase, the reverse reaction speeds up.

the rates of the forward and reverse reactions **become equal**, so the concentracons of the ions **cease to change**.

At this point the system has reached a state of **equilibrium**.

This is what is implied by the **double arrows**, Sometimes an **equal sign (=)** is used in place of the **double arrows**.



when a strong acid such as HC1 is dissolved in water, it reacts so completely to produce H_3O^+ and $C1^-$ that no detectable amount of HC1 molecules remain

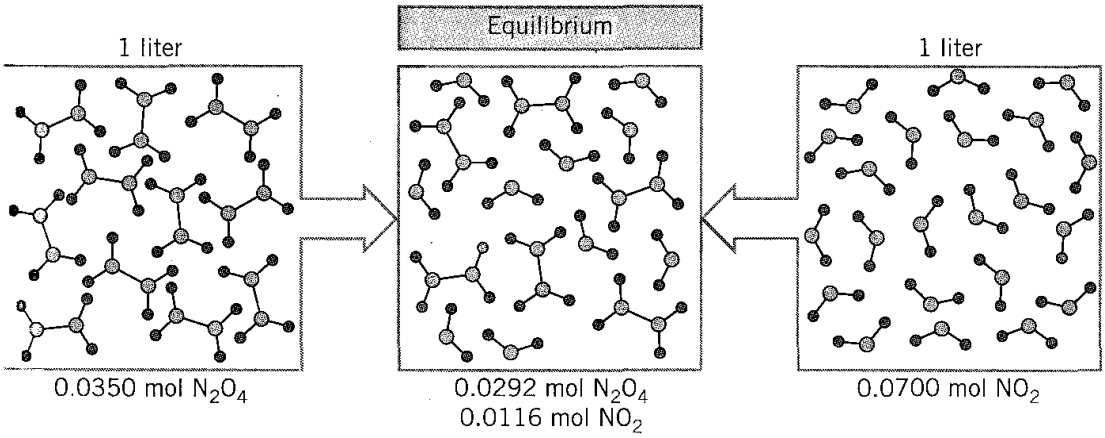
Reaction Reversibility

Acetic acid molecules **are able to react with water** molecules to form the ions, and the ions **are able to react** to form the molecules.

The composition of the equilibrium mixture is found to be independent of whether we begin the reaction from the "reactant side or the "product side."

To illustrate this, let's consider some experiments that we might perform at 25°C on the **gaseous decomposition of N_2O_4 into NO_2**

$$N_2O_4\,(g) = 2\,NO_2(g)$$



In the first 1-liter flask we place 0.0350 mol N_2O_4. Since no NO_2 is present, some N_2O_4 must decompose

When equilibrium is reached, we find the concentration of N_2O_4 has dropped to 0.0292 mol/L and the concentration of NO_2 has become 0.0116 mol/L.

In the second 1-L flask we place 0.0700 mol of NO_2 no N_2O_4 present initially, so NO_2 molecules must combind,

If we measure the concentrations at equilibrium in the second flask, we find, once again, 0.0292 mol/L of N_2O_4 and 0.0116 mol/L of NO_2.

We see here that the **identical composition of the system results whether we begin with pure NO_2 or pure N_2O_4**

We always reach the same equilibrium concentrations whether equilibrium is approached from the forward or reverse direction.

It is in this sense that **chemical reactions are said to be reversible**.

The Equilibrium Law for a Reaction

Gaseous reaction of **H_2(g)** with **I_2(g)** to form **HI (g)**:

$$H_2(g) + I_2(g) = 2HI\ (g)$$

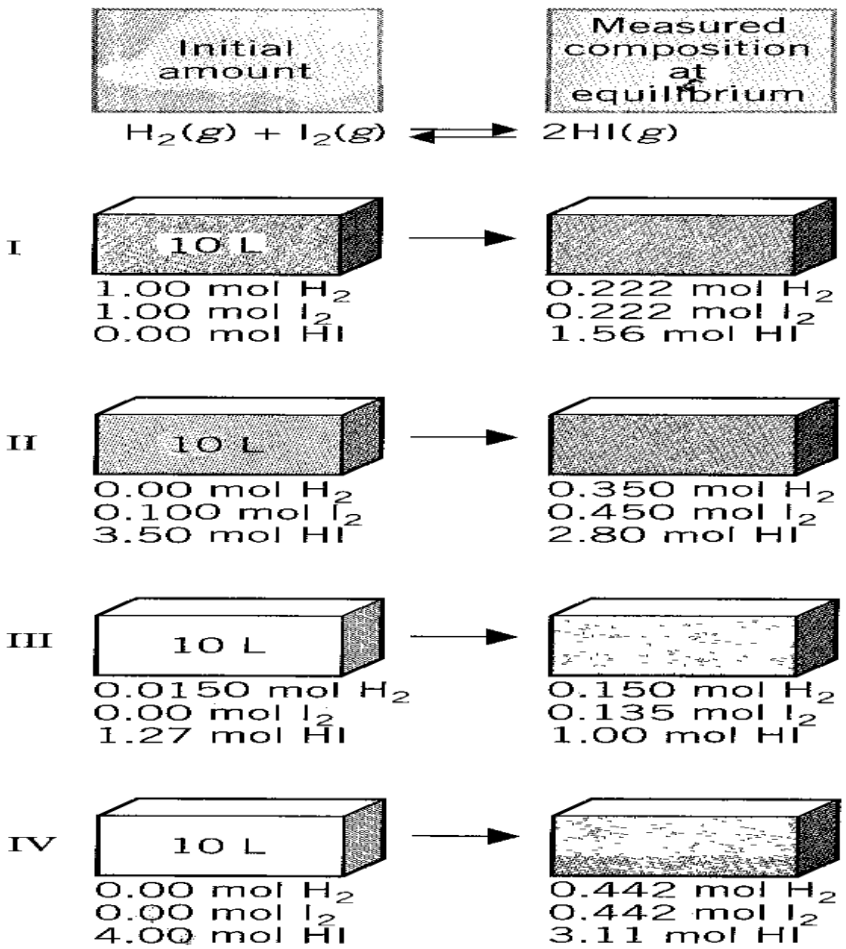
concentrations of H_2, I_2, and HI are different for each experiment.

For each experiment in if we square the molar concentration of HI at equilibrium and then divide this by the product of the equilibrium molar concentrations of H_2 and I_2, **we obtain the same numerical value.**

Table.1: Equilibrium concentrations and Mass Action Expression

Experiment	Equilibrium Concentrations (mol L^{-1}) $[H_2]$	$[I_2]$	$[HI]$	$\frac{[HI]^2}{[H_2][I_2]}$
I	0.0222	0.0222	0.156	$(0.156)^2/(0.0222)(0.0222)$
II	0.0350	0.0450	0.280	$(0.280)^2/(0.0350)(0.0450)$
III	0.0150	0.0135	0.100	$(0.100)^2/(0.0150)(0.0135)$
IV	0.0442	0.0442	0.311	$(0.311)^2/(0.0442)(0.0442)$
				Average

4 experiments to study the equilibrium among H_2, I_2 and HI gases at 440 °C.



$$\frac{[HI]^2}{[H_2][I_2]}$$

is called the **mass action expression**.

The numerical value of the mass action expression is called the reaction quotient, and is often **symbolized by the letter Q**.

the reaction quotient is equal to constant value of 49.5

In fact, if **we repeated the experiments starting with different amounts** of H_2, I_2**,** and HL would still **obtain the same <u>reaction quotient</u>**

provided the systems reached equilibrium and the temperature was 440 °C.

$$\frac{[HI]^2}{[H_2][I_2]} = 49.5 \quad \text{(at 440 °C)}$$

This is the equilibrium law for the system.

It tells us that **for a mixture of the three gases to <u>be at equilibrium</u> at 440 °C, the value of the mass action expression (the reaction quotient) must equal 49.5**.

$$\frac{[HI]^2}{[H_2][I_2]} = K_c = 49.5 \quad \text{(at 440 °C)}$$

For chemical equilibrium to exist in a reaction mixture, **the reaction quotient Q must be equal to the equilibrium constant, K_c**.

Predicting the Equilibrium Law

equilibrium law is that it **can always be predicted from the balanced chemical equation**

For example, **d D + e E = f F+ g G**

where **D, F, F, and G represent chemical formulas** and

d, e, f and g are their coefficients, the **mass action expression** is:

$$\frac{[F]^f[G]^g}{[D]^d[E]^e}$$

The coefficients in the balanced equation are the same as the exponents in the mass action expression.

The condition **for equilibrium** in this reaction is given by the equation:

$$\frac{[F]^f[G]^g}{[D]^d[E]^e} = K_c$$

concentrations that satisfy the equation **are equilibrium concentrations**.

the molar concentrations of the products are always placed in the numerator and those of the reactants appear in the denominator.

Ex.

Write the equilibrium law for the reaction:

$$N_2(g) + 3H_2(g) = 2NH_3(g)$$

$$\frac{[NH_3]^2}{[N_2][H_2]^3} = K_c$$

$$\frac{[HI]^2}{[H_2][I_2]} = K_c$$

its reciprocal is also equal to a constant (let's call it K_c')

$$\frac{[H_2][I_2]}{[HI]^2} = \frac{1}{K_c} = K_c'$$

Manipulating Equations for Chemical Equilibria

Sometimes **it is useful to be able to combine chemical equilibria to obtain the quation for some other reaction of interest**.

i-Changing the Direction of an Equilibrium

$$PCl_3 + Cl_2 \rightleftharpoons PCl_5 \qquad K_c = \frac{[PCl_5]}{[PCl_3][Cl_2]}$$

we obtain

$$PCl_5 \rightleftharpoons PCl_3 + Cl_2 \qquad K_c' = \frac{[PCl_3][Cl_2]}{[PCl_5]}$$

K_c' equals $1/K_c$.

ii-Multiplying the Coefficients by a Factor

Suppose **we multiply the coefficients** of the equation **by 2**:

$$PCl_3 + Cl_2 \rightleftharpoons PCl_5 \qquad K_c = \frac{[PCl_5]}{[PCl_3]\,[Cl_2]}$$

This gives:

$$2PCl_3 + 2Cl_2 \rightleftharpoons 2PCl_5 \qquad K_c'' = \frac{[PCl_5]^2}{[PCl_3]^2[Cl_2]^2}$$

we see that $\mathbf{K_c'' = K_c^2}$

iii-Adding Chemical Equilibria

suppose we add the following two equations.

$$2N_2 + O_2 \rightleftharpoons 2N_2O \qquad K_{c1} = \frac{[N_2O]^2}{[N_2]^2[O_2]}$$

$$2N_2O + 3O_2 \rightleftharpoons 4NO_2 \qquad K_{c2} = \frac{[NO_2]^4}{[N_2O]^2[O_2]^3}$$

$$2N_2 + 4O_2 \rightleftharpoons 4NO_2 \qquad K_{c3} = \frac{[NO_2]^4}{[N_2]^2[O_2]^4}$$

If we multiply the mass action expression for K_{c1} by that for K_{c2}, we obtain the mass action expression for K_{c3}.

$$\frac{\cancel{[N_2O]^2}}{[N_2]^2[O_2]} \times \frac{[NO_2]^4}{\cancel{[N_2O]^2}[O_2]^3} = \frac{[NO_2]^4}{[N_2]^2[O_2]^4}$$

Therefore, $\mathbf{K_{c1} \times K_{c2} = K_{c3}}$

<u>Practice Exercise 2</u>:

At 25 **°C,** K_c =7.0 X 10^{25} for the reaction:

$$\mathbf{2SO_2(g) + O_2(g) = 2SO_3(g)}$$

What is the value of K_c for the reaction:

$$\mathbf{SO_3(g) = SO_2(g) + \tfrac{1}{2}\, O_2(g)}$$

<u>Practice Exercise 3</u>:

At 25 °C, the following reactions have the equilibrium constants noted to the right of their equations.

$\mathbf{2CO(g) + O_2(g) = 2CO_2(g)}$ K_c = **3.3** X 10^{91}

$\mathbf{2H_2(g) + O_2(g) = 2H_2O(g)}$ K_c = **9.1** X 10^{80}

Use these data to calculate K_c for the reaction

$\mathbf{H_2O\ (g) + CO(g) = CO_2(g) + H_2(g)}$